Three-dimensional Yang–Mills: conditional scaling and the physical gap limit

For a **constructed physical** pure Yang–Mills theory in three spacetime (two space) dimensions, with no additional parameters and a fixed square-torus shape, dimensional covariance gives

$$\Delta(L) = E_3 f(x), \qquad E_3 = \hbar^2 c g_{\mathrm{cl}}^2, \qquad x = \hbar g_{\mathrm{cl}}^2 L.$$

The positive finite infinite-volume gap would require $\lim_{x\to\infty} f(x) = C \in (0,\infty)$, together with physical spectral convergence and reconstruction. Dimensional analysis alone proves none of these existence or positivity statements. The constant-mode matrix model and a volume-uniform strong-coupling lattice estimate control different regimes; neither transports itself to that limit. The current-variable kinetic scale $N/(2\pi)$ of Karabali–Nair is not an established full physical gap constant. New research remains suspended; this note maintains the separate continuum obligation and the exact normalization tests.

Physical units and the one-variable form

Use a connection-valued field with $S = (4g_{\mathrm{cl}}^2)^{-1} \int F^a_{\mu\nu} F^a_{\mu\nu}\, d^3x$ and weight $e^{-S_E/\hbar}$. Then $[1/g_{\mathrm{cl}}^2] =$ action length. Put $\lambda_3 = \hbar g_{\mathrm{cl}}^2 = 1/\ell_g$, which is an inverse length, and $E_3 = \hbar c \lambda_3$. The classical coupling by itself is not a mass when physical $\hbar, c$ remain explicit. The mass unit is $\mu = E_3/c^2 = \hbar^2 g_{\mathrm{cl}}^2/c$.

On $T^2_L \times \mathbb{R}$, if the gap depends only on those constants and L, the dimensional theorem makes it E_3 times a function of $x = L/\ell_g$. Vacuum-energy shifts leave a defined excitation gap unchanged; that observation does not establish ultraviolet construction or spectral convergence. The low-dimensional treatment owns the field normalization and solved matrix/circle comparisons.

Constant modes: an exact truncation and a conditional small-volume use

For $SU(2)$ in temporal gauge, take constant spatial components $\vec{a}_1, \vec{a}_2$. With $F_{0i} = c^{-1}\dot{a}_i$ and $F_{12} = \vec{a}_1 \times \vec{a}_2$, their Lagrangian and Hamiltonian are

$$L_0 = \frac{L^2}{2g_{\mathrm{cl}}^2 c}|\dot{\vec{a}}|^2 - \frac{L^2 c}{2g_{\mathrm{cl}}^2}|\vec{a}_1 \times \vec{a}_2|^2,$$

$$H_0 = \frac{g_{\mathrm{cl}}^2 c}{2L^2}\sum_{i=1}^{2}|\vec{p}_i|^2 + \frac{L^2 c}{2g_{\mathrm{cl}}^2}|\vec{a}_1 \times \vec{a}_2|^2, \qquad [a_i^a, p_j^b] = i\hbar\delta_{ij}\delta^{ab}.$$

This is C133 with $D = 2$, $m_{\mathrm{eff}} = L^2/(g_{\mathrm{cl}}^2 c)$ and $g_B^2 = L^2 c/g_{\mathrm{cl}}^2$. Dilation gives the energy unit

$$\hbar^{4/3} g_B^{2/3} m_{\mathrm{eff}}^{-2/3} = \hbar^{4/3} c (g_{\mathrm{cl}}^2)^{1/3} L^{-2/3} = \frac{\hbar c}{L} x^{1/3} = E_3 x^{-2/3}.$$

The **truncated** Gauss-invariant gap is therefore $\Delta_{0,\mathrm{inv}}(L) = \delta^{(2)}_{\mathrm{inv}} E_3 x^{-2/3}$, with $\delta^{(2)}_{\mathrm{inv}} \geq \delta^{(2)}_1 > 0$ from the invariant-sector theorem. The unrestricted scalar or matrix first gap cannot silently replace this physical-sector coefficient. This cross-product proof is $SU(2)$; general $SU(N)$ constant-mode conclusions need their corresponding matrix theorem.

Transport to a full physical finite-volume theory would give $f(x) = \delta^{(2)}_{\mathrm{inv}} x^{-2/3}[1 + o(1)]$ **if** the nonzero-mode coupling, renormalization and sector/domain controls make the relative error vanish. The earlier wording treated that error as already controlled merely by analogy with Lüscher's four-dimensional small-volume method. It is not proved here. The natural scaling parameter $x^{1/3}$ is dimensional bookkeeping, not a remainder estimate. The Feshbach argument states the distinct complement, Schur and effective-Hamiltonian obligations; a finite-matrix gap alone does not satisfy them.

The lattice route and its strong-coupling end

On the periodic square lattice with N_s sites per side, use

$$H_{a,L} = \frac{\hbar c}{a}\left[\frac{g_{\mathrm{lat}}^2}{2}\sum_{\ell}(-\Delta_\ell) + \frac{2}{g_{\mathrm{lat}}^2}\sum_{p}(N - \mathrm{Re}\,\mathrm{tr}\,U_p)\right], \qquad g_{\mathrm{lat}}^2 = \lambda_3 a.$$

Here N is colour rank, $N_s = L/a$ is lattice size and g_{lat} is dimensionless. The finite-lattice theorem gives a unique physical ground state and $\Delta_{a,L} = (\hbar c/a)\delta(g_{\mathrm{lat}}; N_s, G) > 0$. The strong-coupling theorem with spatial $\nu = 2$, $\Lambda_0 = \{0,1\}^2$, two links and one plaquette per site gives

$$\beta = \frac{8N}{g_{\mathrm{lat}}^4 C_2} = \frac{16N^2}{g_{\mathrm{lat}}^4(N^2-1)}, \qquad g_0^{(2)}(N) = \left[\frac{16N^2}{(N^2-1)\beta_*(2,\{0,1\}^2)}\right]^{1/4},$$

$$\Delta_{a,L} \geq \gamma_2 \frac{g_{\mathrm{lat}}^2}{2} C_2 \frac{\hbar c}{a}, \qquad g_{\mathrm{lat}} \geq g_0^{(2)}(N),$$

uniformly in N_s. The unspecified theorem constants β_*, γ_2 retain their source hypotheses; they are not numerical continuum estimates.

At fixed λ_3, L, taking $a \to 0$ sends $g_{\mathrm{lat}} \to 0$ with $x = g_{\mathrm{lat}}^2 N_s$ fixed and $\Delta_{a,L}/E_3 = \delta(g_{\mathrm{lat}}; N_s, G)/g_{\mathrm{lat}}^2$. A finite-volume continuum spectral theorem would need

$$\frac{\delta(g_{\mathrm{lat}}; x/g_{\mathrm{lat}}^2, G)}{g_{\mathrm{lat}}^2} \longrightarrow f(x).$$

For the volume-first lattice route define $\delta_\infty(g_{\mathrm{lat}}) = \liminf_{N_s\to\infty} \delta(g_{\mathrm{lat}}; N_s, G)$ and $c_3(g_{\mathrm{lat}}) = \delta_\infty(g_{\mathrm{lat}})/g_{\mathrm{lat}}^2$. At strong coupling $c_3 \geq \gamma_2 C_2/2$. A finite positive limit $c_3 \to C$ near zero, plus construction, matching of limit orders and surviving physical spectral weight, is the continuum target. Positivity and a common lower bound at **every** intermediate bare coupling form a sufficient optional strengthening, not a necessary all-coupling theorem for the continuum problem. The strong-coupling lower bound does not control the required weak-coupling end.

For compact $U(1)$ the all-coupling Villain gap can coexist with a massless fixed-λ_3 continuum limit. The Villain proof retains normalized measure bounds, its essential monopole suppression and the distinct massive Gaussian scaling trajectory. The former Debye prefactor with dimensionless β/a^3 is not a mass-squared formula in the present physical units and is withdrawn here. These Euclidean source results are not automatically a theorem for the Kogut–Susskind Hamiltonian just defined.

In four dimensions the usual scaling target instead has a transmutation factor: up to its specified higher-order running factors, $\delta_\infty(g)/g^2$ must vanish like $g^{-2}e^{-1/(2b_0 g^2)}$. This is the required trajectory shape if a physical gap survives, not an estimate proved from the strong-coupling end. The three-dimensional energy unit is simpler, but a nonzero spectral coefficient is still an independent dynamical question.

Construction boundary and consequence for STATE

Chandra–Chevyrev–Hairer–Shen construct a renormalized local evolution in **auxiliary stochastic time**, with possible blow-up; the cited result supplies neither global survival, an invariant measure nor OS reconstruction (2024 introduction, passage reading recorded in refinement). Karabali–Nair's $N/(2\pi)$ is a current-variable kinetic scale (1996 equations (28)–(29), passage), not a proved full gap. The cited 2026 outline likewise leaves that proof open; its source boundary is kept in the refinement treatment.

Conditional on the physical construction and limit, the question is whether $f(x)$ has a positive finite limit. A lower bound $f(x) \geq f_- > 0$ for $x \geq x_0$ would supply the positive side only after that construction; one still needs finiteness and an appropriate spectral limit. The matrix proof, conditional small-volume transport, strong lattice estimate and continuum limit

remain separate. This note stays distinct from the solved low-dimensional models because it owns the explicit $\nu = 2$ lattice thresholds, normalization and unresolved field-theoretic limit, not another matrix-gap proof.